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Memory device for performing in-memory computation and operating method thereof

Abstract

A memory device for performing an in-memory computation, comprising a plurality of memory cells each stores a weight value and comprises a transistor and a resistor. A gate of the transistor receives an input voltage, the input voltage indicates an input value. When the transistor operates at a first operating point, the input voltage is equal to a first input voltage, when the transistor operates at a second operating point, the input voltage is equal to a second input voltage. The resistor is connected to a drain and a source of the transistor, when the resistor operates in a first state, the weight value is equal to a first weight value, when the resistor operates in a second state, the weight value is equal to a second weight value. Each of the memory cells performs a product computation of the input value and the weight value.

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Background/Summary

TECHNICAL FIELD

(1) The disclosure relates to a semiconductor device and an operating method thereof, and more particularly, relates to a memory device for performing in-memory computation and an operating method thereof.

BACKGROUND

(2) Artificial intelligence technology and big data processing technology have been widely used in all aspects of daily life. Artificial intelligence technology and big data processing technology rely on a large number of mathematical computations, such as neural network computations.

(3) With the progress of semiconductor technology, in order to increase the computation speed, the above-mentioned mathematical computations may be performed inside the memory device, which is referred to as in-memory computation (IMC). In-memory computations include, for example, data searching, data comparison and sum-of-product computation.

(4) In the sum-of-product computation, the memory cells of the memory device store weight values inside, and the memory cells receive input values from the outside. The memory cells perform sum-of-product computations based on the weight values and the input values, and the memory cells have a processing mechanism to change the weight values.

(5) Those skilled in the art are devoted to improve the process and materials of the memory cells, expecting to be capable of manufacturing the memory cells without additional masks, while the memory cells may still flexibly change the weight values.

SUMMARY

(6) According to an aspect of the disclosure, a memory device for performing an in-memory computation is provided. The memory device comprises a plurality of memory cells, each of the memory cells stores a weight value, and each of the memory cells comprises a transistor and a resistor. The transistor has a gate, a drain and a source, the gate receives an input voltage, the input voltage indicates an input value, when the transistor operates at a first operating point, the input voltage is equal to a first input voltage, when the transistor operates at a second operating point, the input voltage is equal to a second input voltage, the second input voltage is higher than the first input voltage. The resistor is connected to the drain and the source of the transistor, when the resistor operates in a first state, the weight value is equal to a first weight value, when the resistor operates in a second state, the weight value is equal to a second weight value. Each of the memory cells performs a product computation of the input value and the weight value.

(7) According to another aspect of the disclosure, an operating method of a memory device for performing an in-memory computation is provided. The memory device comprises a plurality of memory cells, each of the memory cells comprises a transistor and a resistor, the transistor has a gate, a drain and a source, the resistor is connected to the drain and the source of the transistor, and the operating method comprises the following steps. Receiving an input voltage through the gate of the transistor, the input voltage indicates an input value. Setting the input voltage as a first input voltage, so as to operate the transistor at a first operating point. Setting the input voltage as a second input voltage, so as to operate the transistor at a second operating point, the second input voltage is higher than the first input voltage. Storing a weight value by each of the memory cells. Operating the resistor in a first state, so as to set the weight value as a first weight value. Operating the resistor in a second state, so as to set the weight value as a second weight value. Performing a product computation of the input value and the weight value by each of the memory cells.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic diagram of a memory cell according to an embodiment of the disclosure.

(2) FIG. 2 is a current-voltage curve diagram of the drain current and the gate voltage of the transistor in FIG. 1.

(3) FIG. 3 is a schematic diagram of the memory cell in FIG. 1 performing computations.

(4) FIG. 4 is a schematic diagram of the memory string performing computations, according to an embodiment of the disclosure.

(5) FIG. 5 is a schematic diagram of the memory device performing computations, according to an embodiment of the disclosure.

(6) FIG. 6 is a measurement diagram of the computation results of the memory device in FIG. 5.

(7) In the following detailed description, for purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of the disclosed embodiments. It will be apparent, however, that one or more embodiments may be practiced without these specific details. In other instances, well-known structures and devices are schematically illustrated in order to simplify the drawing.

DETAILED DESCRIPTION

(8) FIG. 1 is a schematic diagram of a memory cell **100-11** according to an embodiment of the disclosure. The memory cell **100-11** includes a transistor **tr11** and a resistor **R11**, the transistor **tr11** is connected to the resistor **R11** in parallel. The transistor **tr11** has a gate **g11**, a source **s11** and a

drain d11. The drain d11 is connected to the first end e1 of the resistor R11, and the source s11 is connected to the second end e2 of the resistor R11. The gate g11 receives the input voltage V1, and the gate voltage Vg of the gate g11 is equal to the input voltage V1.

(9) The drain d11 of the transistor tr11 has a drain current Id, and the memory cell 100-11 has an output current I1. The output current I1 is equal to the sum of the drain current Id and the current of the resistor R11.

(10) In one example, a plurality of memory cells may form a memory string. For example, the memory cell 100-11 and other memory cells (FIG. 1 does not show other memory cells) are connected in series to form a memory string. The source s11 of the transistor tr11 of the memory cell 100-11 may be coupled to the drain of the transistor of a neighboring memory cell. The drain current of the transistor of and the current of the resistor of each of the memory cells in the memory string are summed up to form the output current I1 of the memory string. The computation result of the sum-of-product computation performed by the memory string may be calculated according to the output current I1 (which will be described in detail in embodiments of following paragraphs with FIG. 4).

(11) The resistor R11 is formed of a material having resistive characteristics. In one example, the resistor R11 is, for example, an “anti-fuse”. In other examples, the resistor R11 is, for example, a one-time programmable memory (OTP memory), a multiple-time programmable memory (MTP memory), a Resistive Random Access Memory (ReRAM), or other types of non-volatile memory. Taking the anti-fuse as an example, the resistor R11 implements the characteristics of anti-fuse through the gate oxide breakdown mechanism. The resistor R11 operates in an initial state S_I and a programmed state S_P. The initial state S_I may be referred to as a “first state”, and the programmed state S_P may be referred to as a “second state”.

(12) In the initial state S_I, the resistor R11 is equivalent to a state of being fused, and the resistor R11 is nearly being in a state of turned-OFF. Therefore, the resistor R11 has a first resistance value RH_af which is relatively high. On the other hand, after the programming operation is applied to the resistor R11, the gate oxide layer of the resistor R11 breaks down in response to the applied programming voltage, the state of the resistor R11 is converted to the programmed state S_P, and the resistor R11 is nearly being in a state of turned-ON. Therefore, the resistor R11 has a second resistance value RL_af which is relatively low. The second resistance value RL_af is lower than the first resistance value RH_af, which conforms to the relationship shown in formula (1):

(13) $RH_af > RL_af$ (1)

(14) As shown in Table 1, when the state of the resistor R11 is the initial state S_I, the resistor R11 is in a high-resistance state (high-R state) and has a relatively high first resistance value RH_af. In contrast, when the state of the resistor R11 is the programmed state S_P, the resistor R11 is in a low-resistance state (low-R state) and has a relatively low second resistance value RL_af.

(15) TABLE-US-00001

TABLE 1	State of resistor	high-R state	low-R state	R11	Initial state	S_I	Programmed state	S_P	Resistance value	The first resistance value	The second resistance value
resistor R11	RH_af	RL_af									

(16) In another example, the resistor R11 may also be realized by a “fuse”. The characteristics of the fuse are opposite to those of the anti-fuse described above. When the resistor R11 is a fuse, the resistor R11 in the initial state S_I has a relatively low second resistance value RL_af. After the programming operation is performed to the resistor R11, the resistor R11 is converted to the programmed state S_P, and the resistor R11 has a relatively high first resistance value RH_af.

(17) On the other hand, the transistor tr11 is, for example, an N-type metal oxide semiconductor field effect transistor (N-type MOSFET), which is referred to as “N-type transistor”. The gate g11 of the transistor tr11 receives the input voltage V1. When the input voltage V1 is a relatively high voltage, the channel of the transistor tr11 is substantially turned-on, and the transistor tr11 has a relatively low equivalent resistance. In contrast, when the input voltage V1 is a relatively low

voltage, the channel of the transistor tr11 is substantially turned-off, and the transistor tr11 has a relatively high equivalent resistance value. The relationship between the input voltage V1 and the equivalent resistance value of the transistor tr11 will be described in detail below with reference to FIG. 2.

(18) FIG. 2 is a current-voltage curve diagram of drain current Id and gate voltage Vg of the transistor tr11 in FIG. 1. The gate voltage Vg of the transistor tr11 is equal to the input voltage V1. On the current-voltage curve of the drain current Id and the gate voltage Vg, different operating points of the transistor tr11 may be defined. At the operating point A, the input voltage V1 received by the transistor tr11 is 0V (i.e., the gate voltage Vg of the transistor tr11 is 0V). Since the gate voltage Vg of the transistor tr11 is very low, the channel of the transistor tr11 is almost completely-turned-off, and the equivalent resistance value of the transistor tr11 is very high. Therefore, the drain current Id of the transistor tr11 is very low, and the drain current Id is about 10^{sup.}–12 A.

(19) When the memory cell 100-11 and other memory cells are connected in series to form a memory string, the source of the transistor and the resistor of the last memory cell in the memory string may be connected to a load capacitor CL_1 (not shown in FIG. 1). The drain current Id of the transistor of each of the memory cells in the memory string is added to the current of the resistor to form the output current I1 of the memory string, and the output current I1 charges the load capacitor CL_1.

(20) The drain current Id at the operating point A has a much less current amount (approximately 5×10^{sup.}–13 A), hence the current amount of the output current I1 of the memory string is much less, and the charging time T1 of the load capacitor CL_1 will be very long and exceed a reasonable range for measurement. Therefore, the transistor tr11 of this embodiment is not operated at the operating point A, instead, the input voltage V1 is increased (i.e., the gate voltage Vg is increased), so that the transistor tr11 is slightly turned-on, causing the transistor tr11 to operate at operating point C. The operating point C may be referred to as a “first operating point”. Since the transistor tr11 is slightly turned-on, the transistor tr11 has a first equivalent resistance value RH_tr which is relatively high.

(21) When the transistor tr11 is an N-type transistor, the gate voltage Vg at the operating point C is higher than the threshold voltage Vt of the transistor tr11 multiplied by a first ratio N1. In this embodiment, the first ratio N1 may be “½” or “⅓”, and the gate voltage Vg at the operating point C is 0.9V, as shown in formula (2) or formula (3):

$$V1 = Vg = 0.9V > Vt \times 1 / 2 \quad (2)$$

$$(22) \quad V1 = Vg = 0.9V > Vt \times 1 / 3 \quad (3)$$

(23) At the operating point C, in response to the gate voltage Vg of 0.9V, the current amount of the drain current Id increases to about 5×10^{sup.}–5 A, hence the output current I1 of the memory string has an increased current amount, and the charging time T1 for the drain current Id to charge the load capacitor CL_1 may be within a reasonable range for measurement.

(24) At the operating point C, the input voltage V1 of the transistor tr11 is equal to the first input voltage VL, and the first input voltage VL is 0.9V. The first input voltage VL is higher than the threshold voltage Vt multiplied by the first ratio N1.

(25) On the other hand, the gate voltage Vg of the transistor tr11 is further increased to 1.8V, such that the transistor tr11 operates at the operating point B. The operating point B may be referred to as “the second operating point”. At the operating point B, a greater degree of being-turned-on for the transistor tr11 is achieved, and the equivalent resistance value of the transistor tr11 is reduced to a second equivalent resistance value RL_tr. The second equivalent resistance value RL_tr is lower than the first equivalent resistance value RH_tr at the operating point C, which conforms to the relationship shown in formula (4-1):

$$(26) \quad RH_tr > RL_tr \quad (4 - 1)$$

(27) At the operating point B, the drain current Id of the transistor tr11 has a current amount

increasing up to 5×10^{-4} A. Therefore, the charging time T1 for the output current I1 of the memory string to charge the load capacitor CL_1 is also within a reasonable range for measurement. The input voltage V1 (equal to the gate voltage Vg) of the transistor tr11 is raised to a second input voltage VH. The second input voltage VH is, for example, 1.8V, and the second input voltage VH is higher than the first input voltage VL.

(28) In one example, the relationship of the second equivalent resistance value RL_tr of the transistor tr11 at the operating point B and the first equivalent resistance value RH_tr at the operating point C may be defined according to a second ratio N2. For example, the first equivalent resistance value RH_tr is lower than or equal to a product of the second equivalent resistance value RL_tr and the second ratio N2, wherein the second ratio N2 is e.g., “1000”, as shown in formula (4-2):

$$(29) \quad RH_{tr} \leq RL_{tr} \times N2 \quad (4-2)$$

(30) According to formula (4-1) and formula (4-2), a relationship shown in formula (4-3) may be obtained:

$$(31) \quad RL_{tr} < RH_{tr} \leq RL_{tr} \times N2 \quad (4-3)$$

(32) To sum up, the transistor tr11 has different equivalent resistance values at different operating points B and C on the relationship curve of the drain current Id and gate voltage Vg of FIG. 2.

(33) At operating point C, the gate voltage Vg of the transistor tr11 is 0.9V, and the drain current Id of the transistor tr11 is approximately 5×10^{-5} A, and the transistor tr11 is in a state of high resistance and has a higher first equivalent resistance value RH_tr. On the other hand, at operating point B, the gate voltage Vg of the transistor tr11 is 1.8V, and the drain current Id of the transistor tr11 is approximately 5×10^{-4} A, and the transistor tr11 is in a state of low resistance and has a lower second equivalent resistance value RL_tr.

(34) As shown in Table 2, at the operating point C, the input voltage V1 is equal to the relatively lower first input voltage VL (i.e., the first input voltage VL is equal to the gate voltage Vg of 0.9V), the transistor tr11 is in a high-resistance state having a first equivalent resistance RH_tr. In contrast, at the operating point B, the input voltage V1 is equal to the relatively higher second input voltage VH (i.e., the second input voltage VH is equal to the gate voltage Vg of 1.8V), the transistor tr11 is in a low-resistance state having a second equivalent resistance RL_tr.

(35) TABLE-US-00002 TABLE 2 State of transistor high-R state low-R state tr11 Operating point of Operating point C Operating point B transistor tr11 Input voltage V1 The first input voltage VL The second input voltage VH (equal to the gate voltage (equal to the gate voltage Vg Vg of 0.9 V) of 1.8 V) Drain current Id of 5×10^{-5} A 5×10^{-4} A transistor tr11 Equivalent The first equivalent The second equivalent resistance value of resistance value RH_tr resistance value RL_tr transistor tr11

(36) The first resistance value RH_af and the second resistance value RL_af of the resistor R11 are compared with the first equivalent resistance value RH_tr and the second equivalent resistance value RL_tr of the transistor tr11, the relationship shown in formula (5) may be obtained:

$$(37) \quad RH_{af} > RH_{tr} > RL_{tr} > RL_{af} \quad (5)$$

(38) In another example, the second resistance value RL_af of the resistor R11 is lower than the second equivalent resistance value RL_tr of the transistor tr11, and the relationship shown in formula (6) is obtained:

$$(39) \quad RH_{af} > RH_{tr} > RL_{af} > RL_{tr} \quad (6)$$

(40) FIG. 3 is a schematic diagram of the memory cell 100-11 in FIG. 1 performing computation. The memory cell 100-11 may perform a product computation (or multiplication computation). Referring to FIG. 3 (and refer to FIG. 2 in conjunction), the input voltage V1 of the transistor tr11 may represent the input value X11 of the memory cell 100-11. At the operating point C, the input voltage V1 of the transistor tr11 is equal to the first input voltage VL, which represents that the input value X11 of the memory cell 100-11 is a bit of “1”. At the operating point B, the input

voltage V_1 of the transistor tr_{11} is equal to the second input voltage V_H , which represents that the input value X_{11} of the memory cell **100-11** is a bit of “0”.

(41) On the other hand, the memory cell **100-11** stores a weight value W_{11} , and the weight value W_{11} is used to perform an in-memory computation (IMC) in the memory cell **100-11**. In this embodiment, in-memory computation is a product computation of the input value X_{11} and the weight value W_{11} . The weight value W_{11} may be changed by changing the resistance value of the resistor R_{11} , and the weight value W_{11} may be equal to a “first weight value” or a “second weight value”. In this embodiment, the first weight value is a bit of “1”, and the second weight value is a bit of “0”. The resistor R_{11} has a first resistance value R_{H_af} in the initial state S_I , at this time, the weight value W_{11} is equal to the first weight value (i.e., “1”). On the other hand, the resistor R_{11} has the second resistance value R_{L_af} in the programmed state S_P , at this time, the weight value W_{11} is equal to the second weight value (i.e., “0”).

(42) The value of the weight value (i.e., the first weight value or the second weight value) may be changed by changing the state of the resistor R_{11} (i.e., the initial state S_I or the programmed state S_P) so that the resistor R_{11} has different resistance values. Therefore, there is no need to dispose additional masks in the manufacturing process of the memory cell **100-11** to change the value of the weight value W_{11} . The memory cell **100-11** may adopt manufacturing process of logic circuit, such that the memory cell **100-11** may realize the in-memory computation.

(43) Table 3 shows four conditions of the product computation of the input value X_{11} of the memory cell **100-11** and the weight value W_{11} . In the first condition (cond-1), the input voltage V_1 of the transistor tr_{11} is equal to the second input voltage V_H , and the resistor R_{11} is in the programmed state S_P , indicating that the input value X_{11} is “0” and the weight value W_{11} is “0”. The transistor tr_{11} has a second equivalent resistance value R_{L_tr} , and the resistor R_{11} has a second resistance value R_{L_af} , and the total equivalent resistance value of the memory cell **100-11** is equal to a parallel computation of the second equivalent resistance value R_{L_tr} and the second resistance value R_{L_af} . According to formula (5), the second equivalent resistance value R_{L_tr} is higher than the second resistance value R_{L_af} , hence, the total equivalent resistance value of the memory cell **100-11** is substantially equal to the second resistance value R_{L_af} .

(44) In the second condition (cond-2), the input voltage V_1 of the transistor tr_{11} is equal to the first input voltage V_L , and the resistor R_{11} is in the programmed state S_P , indicating that the input value X_{11} is “1” and the weight value W_{11} is “0”. The transistor tr_{11} has a first equivalent resistance value R_{H_tr} , and the resistor R_{11} has a second resistance value R_{L_af} , the total equivalent resistance value of the memory cell **100-11** is equal to the parallel computation of the first equivalent resistance value R_{H_tr} and the second resistance value R_{L_af} . According to formula (5), the first equivalent resistance value R_{H_tr} is higher than the second resistance value R_{L_af} , hence the total equivalent resistance value of the memory cell **100-11** is substantially equal to the second resistance value R_{L_af} .

(45) In the third condition (cond-3), the input voltage V_1 of the transistor tr_{11} is equal to the second input voltage V_H , and the resistance R_{11} is in the initial state S_I , indicating that the input value X_{11} is “0” and the weight value W_{11} is “1”. The transistor tr_{11} has a second equivalent resistance value R_{L_tr} , and the resistor R_{11} has a first resistance value R_{H_af} , the total equivalent resistance value of the memory cell **100-11** is equal to a parallel computation of the second equivalent resistance value R_{L_tr} and the first resistance value R_{H_af} . According to formula (5), the second equivalent resistance value R_{L_tr} is lower than the first resistance value R_{H_af} , hence the total equivalent resistance value of the memory cell **100-11** is substantially equal to the second equivalent resistance value R_{L_tr} .

(46) In the fourth condition (cond-4), the input voltage V_1 of the transistor tr_{11} is equal to the first input voltage V_L , and the resistance R_{11} is in the initial state S_I , indicating that the input value X_{11} is “1” and the weight value W_{11} is “1”. The transistor tr_{11} has a first equivalent resistance value R_{H_tr} , and the resistor R_{11} has a first resistance value R_{H_af} , the total equivalent resistance

value of the memory cell **100-11** is equal to a parallel computation of the first equivalent resistance value RH_{tr} and the first resistance value RH_{af} . According to formula (5), the first equivalent resistance value RH_{tr} is lower than the first resistance value RH_{af} , hence the total equivalent resistance value of the memory cell **100-11** is substantially equal to the first equivalent resistance value RH_{tr} .

(47) TABLE-US-00003 TABLE 3 State of resistor R11 State of resistor R11 Programmed state S_P Initial state S_I Weight value $W11 = "0"$ Weight value $W11 = "1"$ Product computation of the (equal to the second weight (equal to the first weight memory cell 100-1 value) value) (cond-1) (cond-3) Input voltage $V1 = V_H$ $RL_{af} // RL_{tr} \cong RL_{af}$ $RH_{af} // RL_{tr} \cong RL_{tr}$ Input value $X11 = "0"$ Product result $Y11 = "0"$ Product result $Y11 = "0"$ (cond-2) (cond-4) Input voltage $V1 = V_L$ $RL_{af} // RH_{tr} \cong RL_{af}$ $RH_{af} // RH_{tr} \cong RH_{tr}$ Input value $X11 = "1"$ Product result $Y11 = "0"$ Product result $Y11 = "1"$

(48) The product result $Y11$ shown in Table 3 is a result of product computation of the input value $X11$ and the weight value $W11$ of the memory cell **100-11**. In one example, the total equivalent resistance value of the memory cell **100-11** may reflect the product result $Y11$.

(49) The output current $I1$ of the memory cell **100-11** is equal to the sum of the drain current I_d of the transistor $tr11$ and the current of the resistor $R11$. In another example, the output current $I1$ may also reflect the product result $Y11$ of the memory cell **100-11**. When the memory cell **100-11** and other memory cells are connect in series to form a memory string, and the memory string is connected to a load capacitor CL_1 (not shown in FIG. 3), the output current of the memory cell **100-11** and the output currents of other memory cells are summed up to form the output current $I1$. The output current $I1$ charges the load capacitor CL_1 , and the charging time $T1$ of the load capacitor CL_1 is related to the output current $I1$. Therefore, the charging time $T1$ may also reflect a sum-of-product result. The sum-of-product result of the memory string is a sum of the product result $Y11$ of each of the memory cells (which will be described in detail in embodiments in following paragraphs with FIG. 4).

(50) When the data width of the input value $X11$ and the weight value $W11$ is one bit, the product computation of the input value $X11$ and the weight value $W11$ is equivalent to a logic "AND" operation.

(51) In the first condition (cond-1) in Table 3, the input voltage $V1$ is equal to the second input voltage V_H , and the total equivalent resistance value of the memory cell **100-11** is substantially equal to the second resistance value RL_{af} . Therefore, the output current $I1$ is related to the input voltage $V1$ divided by the second resistance value RL_{af} . Since the input voltage $V1$ is equal to the second input voltage V_H which is relatively high, the second resistance value RL_{af} is a lower resistance value, therefore, the output current $I1$ has a higher current value, resulting in a shorter charging time $T1$. In the first condition (cond-1), the input value $X11$ is "0" and the weight value $W11$ is "0", hence the product result $Y11$ is "0", and the product result $Y11$ is reflected by the shorter charging time $T1$.

(52) In the second condition (cond-2), the input voltage $V1$ is equal to the lower first input voltage V_L , and the total equivalent resistance value of the memory cell **100-11** is substantially equal to the lower second resistance value RL_{af} . Therefore, the current value of the output current $I1$ in the second condition (cond-2) is slightly lower than that in the first condition (cond-1), resulting in a slight increase in the charging time $T1$. In the second condition (cond-2), the input value $X11$ is "1" and the weight value $W11$ is "0", hence the product result $Y11$ is "0".

(53) In the third condition (cond-3), the input voltage $V1$ is equal to the higher second input voltage V_H , and the total equivalent resistance value of the memory cell **100-11** is substantially equal to the lower second equivalent resistance value RL_{tr} , therefore, the current value of the output current $I1$ in the third condition (cond-3) is slightly increased as compared to that of the second condition (cond-2), resulting in a decrease of the charging time $T1$. In the third condition (cond-3), the input value $X11$ is "0" and the weight value $W11$ is "1", hence the product result $Y11$ is "0".

(54) In the fourth condition (cond-4), the input voltage **V1** is equal to the lower first input voltage **VL**, and the total equivalent resistance value of the memory cell **100-11** is substantially equal to the higher first equivalent resistance value **RH_{tr}**, therefore, the current value of the output current **I1** in the fourth condition (cond-4) is greatly lower than that in the third condition (cond3), resulting in a greatly increase in the charging time **T1**. In the fourth condition (cond-4), the input value **X11** is “1” and the weight value **W11** is “1”, hence the product result **Y11** is “i”, and the product result **Y11** is reflected by the greatly increased charging time **T1**.

(55) FIG. 4 is a schematic diagram of the memory string **10-1** performing computations according to an embodiment of the disclosure. As afore-mentioned, the memory cell **100-11** and other memory cells are connected in series to form a memory string. The source **s11** of transistor **tr11** of memory cell **100-11** may be coupled to the drain of transistor of neighboring memory cell. Drain current of transistor and current of resistor of each memory cell in the memory string are summed up to form the output current **I1**. The computation result of sum-of-product computation performed by the memory string may be calculated according to the output current **I1**. More particularly, as shown in FIG. 4, the memory string **10-1** includes **N** memory cells **100-11~100-1N**. Each of the memory cells **100-11~100-1N** is the same as the memory cell **100-11** in FIG. 1. For example, the memory cell **100-12** includes a transistor **tr12** and a resistor **R12** connected in parallel, the memory cell **100-13** includes a transistor **tr13** and a resistor **R13** connected in parallel, and so on. The memory string **10-1** is used to perform a sum-of-product computation, which sums up the respective product results **Y11~Y1N** of the memory cells **100-11~100-1N**, so as to obtain a sum-of-product result **YS₁**.

(56) The memory cells **100-11~100-1N** are connected in series. For example, the source **s11** of the transistor **tr11** of the memory cell **100-11** is connected to the drain **d12** of the transistor **tr12** of the next neighboring memory cell **100-12**, and the second end **e2** of the resistor **R11** is connected to the first end **e3** of the resistor **R12**. Similarly, the source **s12** of the transistor **tr12** of the memory cell **100-12** is connected to the drain **d13** of the transistor **tr13** of the next neighboring memory cell **100-13**, and the second end **e4** of the resistor **R12** is connected to the first end **e5** of the resistor **R13**, and so forth.

(57) The gate **g11** of the transistor **tr11** is connected to the first signal line **WL1** to receive the input voltage **V1**, and the gate **g12** of the next neighboring transistor **tr12** is connected to the first signal line **WL2** to receive the input voltage **V2**, and so on, the gate **g1N** of the **N**th transistor **tr1N** (i.e., the last transistor in the memory string **10-1**) is connected to the first signal line **WLN** to receive the input voltage **VN**. These first signal lines **WL1~WLN** are, for example, word lines.

(58) For the first memory cell **100-11**, the drain **d11** of the transistor **tr11** and the first end **e1** of the resistor **R11** are connected to the second signal line **BL1**, and the transistor **tr11** and the resistor **R11** receive the read voltage **Vread** through the second signal line **BL1**. The second signal line **BL1** is, for example, a bit line.

(59) For the **N**th memory cell **100-1N** (i.e., the last memory cell in the memory string **10-1**), the source **s1N** of the transistor **tr1N** and the second end **e2N** of the resistor **R1N** are connected to the third signal line **SL1**. The memory cell **100-1N** transmits the output current **I1** through the third signal line **SL1**. The third signal line **SL1** is, for example, a source line. The output current **I1** and the output voltage **Vout1** of the memory string **10-1** are transmitted to a sensing line **SS1**.

(60) The first end **eC1** of the load capacitor **CL₁** is connected to the third signal line **SL1** to receive the output current **I1**. That is, the source **s1N** of the transistor **tr1N** of the last memory cell **100-1N** in the memory string **10-1** and the second end **e2N** of the resistor **R1N** are connected to the first end **eC1** of the load capacitor **CL₁**, and the load capacitor **CL₁** receives the output current **I1** of the memory string **10-1**. The output current **I1** charges the load capacitor **CL₁**. The charging time **T1** of the load capacitor **CL₁** reflects the sum-of-product result **YS₁** of the memory string **10-1**.

(61) The memory string **10-1** cooperates with the comparator **200**. The input end **201** of the

comparator **200** is connected to the first end eC1 of the load capacitor CL_1 through the sense line SS1. The input end **201** of the comparator **200** receives the voltage of the first end eC1 of the load capacitor CL_1 (which is equal to an output voltage Vout1). Another input end **202** of the comparator **200** receives a reference voltage Vref, and the comparator **200** compares the output voltage Vout1 with the reference voltage Vref.

(62) At a starting time point t0, the memory string **10-1** starts to operate, and the output current I1 starts to charge the load capacitor CL_1. During the charging process, the voltage at the first end eC1 of the load capacitor CL_1 (which is equal to the output voltage Vout1) continues to increase. Then, at an ending time point t1, the voltage of the first end eC1 of the load capacitor CL_1 reaches the reference voltage Vref (i.e., the output voltage Vout1 is equal to the reference voltage Vref), indicating that the charging-level of the load capacitor CL_1 has met a predefined condition. The time difference between the ending time point t1 and the starting time point t0 may be defined as the charging time T1 of the load capacitor CL_1. The charging time T1 reflects the sum-of-product result YS_1 of memory string **10-1**.

(63) FIG. 5 is a schematic diagram of the memory device **1000** performing computations according to an embodiment of the disclosure. As shown in FIG. 5, the memory device **1000** includes M memory strings **10-1~10-M**, and these memory strings **10-1~10-M** form a memory array, with M being “24”, for example. Each of the memory strings **10-1~10-M** is the same as the memory string **10-1** in FIG. 4. For example, the memory string **10-2** includes N memory cells **100-21~100-2N**. N is “32”, for example. The memory strings **10-1~10-M** are arranged in parallel. The total number of all memory cells **100-11~100-MN** included in the memory device **1000** is M times N, for example, “32” times “24”, which is equal to “768”.

(64) The memory cell **100-11** of the memory string **10-1**, the memory cell **100-21** of the memory string **10-2**, . . . , and the memory cell **100-M1** of the memory string **10-M** are jointly connected to the second signal Line BL1 to receive the read voltage Vread.

(65) The memory cell **100-1N** of the memory string **10-1**, the memory cell **100-2N** of the memory string **10-2**, . . . , and the memory cell **100-MN** of the memory string **10-M** are jointly connected to the sensing line SS1, and connected to the first end eC1 of the load capacitor CL_1 through the sensing line SS1. Memory string **10-1** generates output current I1, memory string **10-2** generates output current I2, and so on, memory string **10-M** generates output current IM. The output currents I1~IM are transmitted to the sensing line SS1 and summed to form a total output current IS_1. The total output current IS_1 charges the load capacitor CL_1.

(66) The memory device **1000** is used to perform a sum-of-product computation, and sum up the sum-of-product results YS_1~YS_M of the memory strings **10-1~10-M** to obtain a total sum-of-product result YST. The charging time TS1 for the total output current IS_1 to charge the load capacitance CL_1 may reflect the total sum-of-product result YST.

(67) In this embodiment, the component parameters and operating parameters of the memory device **1000** are listed as follows. The effective length of the channel under the oxide layer of respective transistors tr11~trMN of the memory cells **100-11~100-MN** is e.g., 0.18 μm , and the effective width is e.g., 0.3 μm . The first resistance value RH_af of respective resistors R11~RMN of the memory cells **100-11~100-MN** is e.g., 100M Ω , and the second resistance value RL_af is e.g., 1K Ω . The read voltage Vread received by the memory device **1000** through the second signal line BL1 is, for example, 0.5V. The capacitance value of the load capacitor CL_1 is e.g., 1 pf. The reference voltage Vref of the comparator **200** is e.g., 0.3V. When the voltage of the first end eC1 of the load capacitor CL_1 reaches 0.3V, the ending time point t1 is recorded, and the charging time T1 of the load capacitor CL_1 is computed accordingly.

(68) FIG. 6 is a measurement diagram of the computation results of the memory device **1000** in FIG. 5. The value on the horizontal axis of the measurement diagram in FIG. 6 represents the total sum-of-product result YST of sum-of-product computation performed by the memory device **1000**, and the value on the vertical axis represents the charging time T1 of the load capacitor CL_1.

(69) The measurement diagram includes a plurality of measurement points, and these measurement points are substantially divided into two types, including: a plurality of first measurement points **m1** of a first type and a plurality of second measurement points **m2** of a second type. Each of the first measurement points **m1** is shown as a “thick black dot” and each of the second measurement points **m2** is shown as a “short cross lines”. When the weight values $W_{11} \sim W_{MN}$ of the memory cells **100-11~100-MN** of the memory device **1000** are all “1”, the computation results of the memory device **1000** are displayed as the first measurement points **m1**. For example, one first measurement point **m1-1** indicates that: the total sum-of-product result **YST** of the memory device **1000** is equal to “700”, and the charging time **T1** of the load capacitor **CL_1** is equal to 0.8 μ s.

(70) On the other hand, when half of the weight values $W_{11} \sim W_{MN}$ of the memory cells **100-11~100-MN** of the memory device **1000** are “1”, the computation results of the memory device **1000** are displayed as the second measurement points **m2**.

(71) The first measurement points **m1** and the second measurement points **m2** substantially show a linear distribution, indicating that the charging time **T1** of the load capacitor **CL_1** is related to the total sum-of-product result **YST** of the memory device **1000**.

(72) It will be apparent to those skilled in the art that various modifications and variations may be made to the disclosed embodiments. It is intended that the specification and examples be considered as exemplary only, with a true scope of the disclosure being indicated by the following claims and their equivalents.

Claims

1. A memory device, for performing an in-memory computation, the memory device comprising: a plurality of memory cells, forming a plurality of memory strings each performing a sum-of-product computation that is a sum operation for a product computation of each of the memory cells, each of the memory cells stores a weight value, and each of the memory cells comprising: a transistor, having a gate, a drain and a source, the gate receives an input voltage, the input voltage indicates an input value, when the transistor operates at a first operating point, the input voltage is equal to a first input voltage, when the transistor operates at a second operating point, the input voltage is equal to a second input voltage, the second input voltage is higher than the first input voltage; and a resistor, connected to the drain and the source of the transistor, when the resistor operates in a first state, the weight value is equal to a first weight value, when the resistor operates in a second state, the weight value is equal to a second weight value, wherein, each of the memory cells performs the product computation of the input value and the weight value, and each of the memory strings is respectively connected to a load capacitor and generates an output current, and a charging time for the output current to charge the load capacitor reflects a sum-of-product result of the sum-of-product computation.
2. The memory device according to claim 1, wherein, the first weight value is a bit of “1”, the second weight value is a bit of “0”, when the input voltage is equal to the first input voltage, the input value is a bit of “1”, when the input voltage is equal to the second input voltage, the input value is a bit of “0”.
3. The memory device according to claim 2, wherein, the first input voltage is higher than a product of a threshold voltage of the transistor and a first ratio.
4. The memory device according to claim 1, wherein, the resistor is an anti-fuse, the first state is an initial state, and the second state is a programmed state, when the resistor operates in the initial state, the resistor has a first resistance value, when the resistor operates in the programmed state, the resistor has a second resistance value, and the first resistance value is higher than the second resistance value.
5. The memory device according to claim 4, wherein, when the transistor operates at the first operating point, the transistor has a first equivalent resistance value, when the transistor operates at

the second operating point, the transistor has a second equivalent resistance value, the first equivalent resistance value is higher than the second equivalent resistance value.

6. The memory device according to claim 5, wherein, the first equivalent resistance value is lower than or equal to a product of the second equivalent resistance value and a second ratio.

7. The memory device according to claim 5, wherein: the first resistance value is higher than the first equivalent resistance value; the first equivalent resistance value is higher than the second equivalent resistance value; and the second equivalent resistance value is higher than the second resistance value.

8. The memory device according to claim 5, wherein: the first resistance value is higher than the first equivalent resistance value; the first equivalent resistance value is higher than the second resistance value; and the second resistance value is higher than the second equivalent resistance value.

9. An operating method of a memory device, for performing an in-memory computation, the memory device comprises a plurality of memory cells which form a plurality of memory strings each performing a sum-of-product computation that is a sum operation for a product computation of each of the memory cells, each of the memory cells comprises a transistor and a resistor, the transistor has a gate, a drain and a source, the resistor is connected to the drain and the source of the transistor, and the operating method comprising: receiving an input voltage through the gate of the transistor, the input voltage indicates an input value; setting the input voltage as a first input voltage, so as to operate the transistor at a first operating point; setting the input voltage as a second input voltage, so as to operate the transistor at a second operating point, the second input voltage is higher than the first input voltage; storing a weight value by each of the memory cells; operating the resistor in a first state, so as to set the weight value as a first weight value; operating the resistor in a second state, so as to set the weight value as a second weight value; and performing the product computation of the input value and the weight value by each of the memory cells, wherein each of the memory strings is respectively connected to a load capacitor and generates an output current, and a charging time for the output current to charge the load capacitor reflects a sum-of-product result of the sum-of-product computation.

10. The operating method according to claim 9, wherein, the first weight value is a bit of "1", the second weight value is a bit of "0", when the input voltage is equal to the first input voltage, the input value is a bit of "1", when the input voltage is equal to the second input voltage, the input value is a bit of "0".

11. The operating method according to claim 10, wherein, the first input voltage is higher than a product of a threshold voltage of the transistor and a first ratio.

12. The operating method according to claim 9, wherein, the resistor is an anti-fuse, the first state is an initial state, and the second state is a programmed state, when the resistor operates in the initial state, the resistor has a first resistance value, when the resistor operates in the programmed state, the resistor has a second resistance value, and the first resistance value is higher than the second resistance value.

13. The operating method according to claim 12, wherein, when the transistor operates at the first operating point, the transistor has a first equivalent resistance value, when the transistor operates at the second operating point, the transistor has a second equivalent resistance value, the first equivalent resistance value is higher than the second equivalent resistance value.

14. The operating method according to claim 13, wherein, the first equivalent resistance value is lower than or equal to a product of the second equivalent resistance value and a second ratio.

15. The operating method according to claim 13, wherein: the first resistance value is higher than the first equivalent resistance value; the first equivalent resistance value is higher than the second equivalent resistance value; and the second equivalent resistance value is higher than the second resistance value.

16. The operating method according to claim 13, wherein: the first resistance value is higher than

the first equivalent resistance value; the first equivalent resistance value is higher than the second resistance value; and the second resistance value is higher than the second equivalent resistance value.
